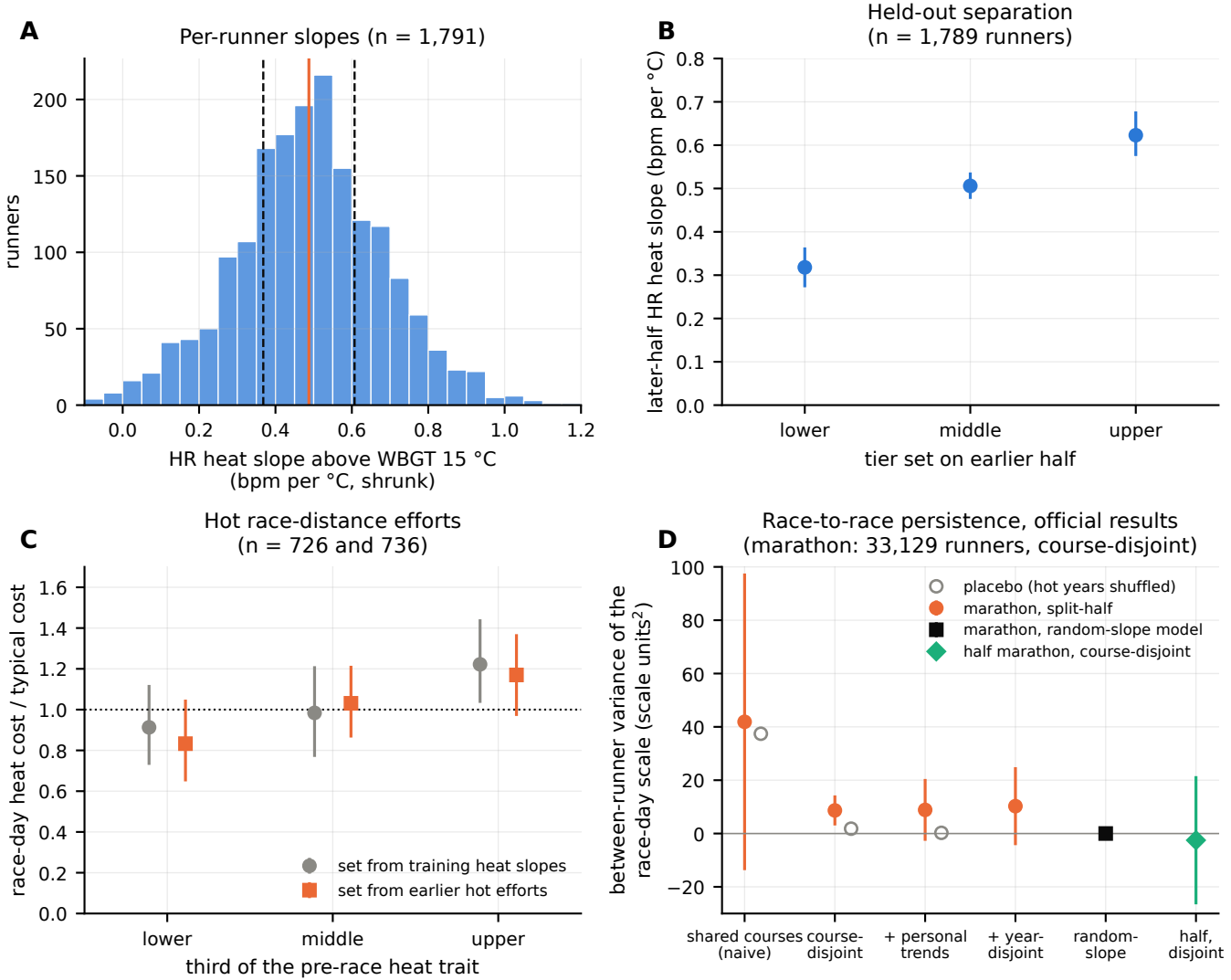
A
Per-runner slopes (n = 1,791)
runners
HR heat slope above WBGT 15 °C
(bpm per °C, shrunk)
B
Held-out separation
(n = 1,789 runners)
later-half HR heat slope (bpm per °C)
lower
middle
upper
tier set on earlier half
C
Hot race-distance efforts
(n = 726 and 736)
race-day heat cost / typical cost
set from training heat slopes
set from earlier hot efforts
third of the pre-race heat trait
D
Race-to-race persistence, official results
(marathon: 33,129 runners, course-disjoint)
between-runner variance of the
race-day scale (scale units²)
placebo (hot years shuffled)
marathon, split-half
marathon, random-slope model
half marathon, course-disjoint
shared courses (naive)
course-disjoint
+ personal trends
+ year-disjoint
random-slope
half, disjoint